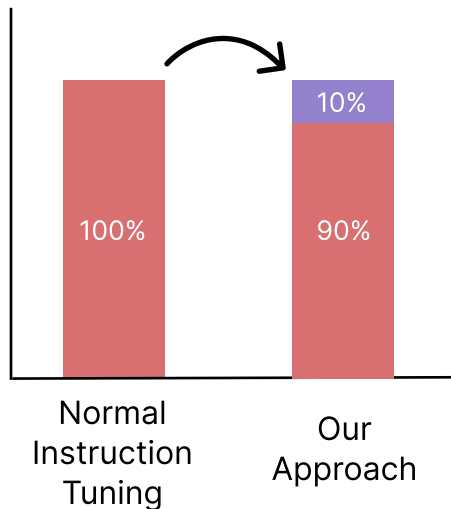


## Idea

Human/LLM-generated instruction-following data

Program-generated multi-hop derivations



## Example of Program-generated Data Point

**Prompt** c3 is ruby. c3 is north. c7 is olive.  
If c3 is north and c3 is ruby, then c9 is teal.  
If c9 is teal, then c9 is west.  
If c9 is west and c7 is olive, then c4 is amber.  
What is c4?



**Latent Proof** Rc3, Nc3  $\Rightarrow$  Tc9  $\Rightarrow$  Wc9, with Oc7  $\Rightarrow$  Ac4



### Formal Target

```
<premises>
R c3 ; N c3 ; O c7
N c3 & R c3  $\rightarrow$  T c9
T c9  $\rightarrow$  W c9
W c9 & O c7  $\rightarrow$  A c4
</premises>
<proof> R c3 ; premise
N c3 ; premise
N c3 & R c3 ; &l,2,1
T c9 ;  $\rightarrow$ E,3
W c9 ;  $\rightarrow$ E,4
O c7 ; premise
W c9 & O c7 ; &l,5,6
A c4 ;  $\rightarrow$ E,7 </proof>
<answer> amber </answer>
```

### English Target

```
<think>
c3 is ruby and c3 is north.
Since c3 is north and c3 is
ruby, c9 is teal.
Since c9 is teal, c9 is west.
c7 is olive.
Since c9 is west and c7 is
olive, c4 is amber.
Therefore, c4 is amber.
</think>
<answer> amber </answer>
```

## Downstream Performance

Baseline Formal 5% English 10%

